🔷 16.5 Probability Evolution and PAC-Based Learning

16.5.1 Fokker–Planck Closure

$$\partial_t P(x,t) = -\nabla \cdot (P F) + \frac{\sigma^2}{2}\nabla^2 P$$

subject to:

- $P \in C^2(\Omega)$
- $P(x,t) \ge \varepsilon > 0$

Thus:

the probability density evolves consistently with the system dynamics.

16.5.2 Mass Conservation

$$\int_{\Omega} P(x,t)\,dx = 1$$

Thus:

total probability is preserved over time.

16.5.3 Log-Stability

$$|\nabla \ln P| = \frac{|\nabla P|}{P} \le \frac{C}{\varepsilon}$$

Thus:

logarithmic gradients remain bounded, preventing singularities.

16.5.4 KL Divergence

$$\Gamma = \int_{\Omega} P(x,t)\,\log\frac{P(x,t)}{P_0(x)}\,dx$$

$$\Gamma \ge 0$$

16.5.5 PAC Definition

$$PAC = \frac{1}{1 + e^{-\Gamma}}$$

Thus:
$PAC \in (\varepsilon, 1 - \varepsilon)$

representing a normalized decision confidence measure.

16.5.6 PAC Stability

$$|PAC_{k+1} - PAC_k| \le L\,dt$$

Thus:

PAC evolves smoothly over time.

16.5.7 Learning Injection into Dynamics

$$F \supset \alpha(PAC_k)\nabla \ln P_k$$

with:
$$0 < \alpha(PAC) \le \alpha_{\max}$$

Thus:

probabilistic learning directly influences system dynamics.

16.5.8 PAC–Graph Coupling

$$w_{ij} \propto PAC_i \cdot PAC_j$$

Thus:

learning signals shape the adaptive network structure.

16.5.9 Invariant Distribution

$$\partial_t P = 0 \;\Rightarrow\; P = \pi(x)$$

Thus:

a steady-state distribution exists.

16.5.10 Integration with System Components

- From Section 16.2: F depends on P
- From Section 16.3: KL divergence contributes to Lyapunov functional
- From Section 16.4: PAC modulates

network coupling

Thus:

probability, learning, and dynamics are fully integrated.

Closure Statement (Part V)

$$\boxed{
\text{The system integrates probability evolution, learning, and control into a unified adaptive framework.}
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